



Sheaf-Regularized Graph Neural Network for Multi-Agent Coordination

Standard GNN Aggregation. In a standard graph neural network, each agent i updates its embedding by aggregating neighbor messages symmetrically:

$$\mathbf{h}'_i = \text{AGG}(\{\mathbf{W}\mathbf{h}_j : j \in \mathcal{N}(i)\}) \quad (1)$$

where $\mathbf{W}$ is a shared weight matrix applied identically to every neighbor. This treats all pairwise interactions as structurally equivalent and provides no mechanism for agents to form mutually consistent representations of their shared interactions.

Sheaf Structure. A cellular sheaf $\mathcal{F}$ over the communication graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ assigns:

- A stalk $\mathcal{F}(\mathbf{v}_i) \cong \mathbb{R}^d$ at each agent node i
- A stalk $\mathcal{F}(\mathbf{e}_{ij}) \cong \mathbb{R}^k$ at each edge (i, j) , where $k = d/2$
- *Restriction maps* $\mathbf{F}_{i \rightarrow j} : \mathbb{R}^d \rightarrow \mathbb{R}^k$ projecting each agent's stalk onto the shared edge stalk

The restriction maps are learned as functions of the edge features $\mathbf{e}_{ij}$ via a shared MLP:

$$\mathbf{F}_{i \rightarrow j} = \text{MLP}_\theta(\mathbf{e}_{ij}) \in \mathbb{R}^{k \times d} \quad (2)$$

Sheaf Convolution. The sheaf layer replaces standard aggregation with restriction-map-projected aggregation:

$$\mathbf{h}'_i = \phi \left(\mathbf{U}\mathbf{h}_i + \frac{1}{|\mathcal{N}(i)|} \sum_{j \in \mathcal{N}(i)} \mathbf{F}_{j \rightarrow i} \mathbf{h}_j \right) \quad (3)$$

where $\mathbf{U}$ is a residual projection and ϕ is a nonlinearity. Each neighbor's embedding is projected through a restriction map specific to that edge before aggregation, allowing the network to extract only the component of each neighbor's state that is relevant to the particular pairwise interaction.

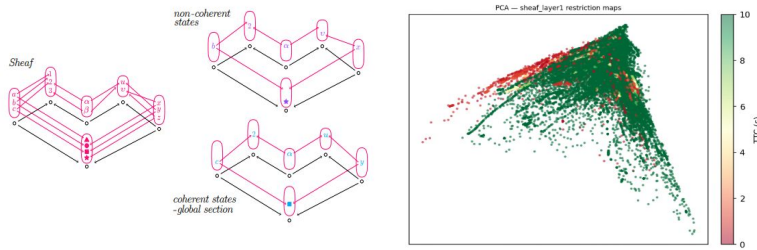
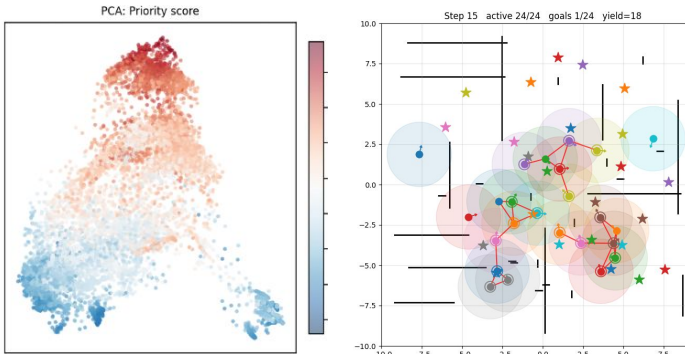
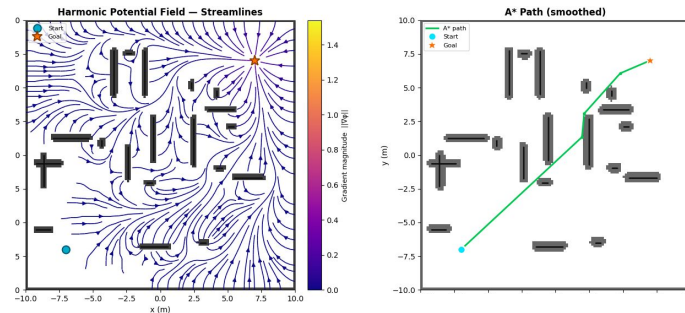
Sheaf Laplacian and Inconsistency Loss. The *sheaf coboundary operator* $\delta_{\mathcal{F}}$ measures disagreement between neighboring agents' projected representations. For an assignment of embeddings $\mathbf{x} = (\mathbf{h}_1, \dots, \mathbf{h}_N)$, the sheaf Laplacian quadratic form is:

$$\mathbf{x}^\top \Delta_{\mathcal{F}} \mathbf{x} = \sum_{(i,j) \in \mathcal{E}} \|\mathbf{F}_{i \rightarrow j} \mathbf{h}_i - \mathbf{F}_{j \rightarrow i} \mathbf{h}_j\|^2 \quad (4)$$

This is minimized when the projected embeddings agree across every edge. We add this as an auxiliary regularization term to the PPO objective:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{PPO}} + \lambda \cdot \mathbf{x}^\top \Delta_{\mathcal{F}} \mathbf{x} \quad (5)$$

Navigation Function Comparison — walls_obstacles



Sheaf GNN vs. Harmonic Baseline 100 episodes, $N = 45$ agents

Metric	Harmonic	Sheaf GNN	Improvement
Goal reach rate	0.930	0.974	+4.7%
Collision rate	0.062	0.021	-66.1%
Avg. goals reached	26.95	28.41	+5.4%
Avg. collisions	2.02	0.65	-67.8%
Avg. steps	252.7	232.2	-8.1%

GAT GNN vs. Harmonic Baseline 100 episodes, $N = 40$ agents

Metric	Harmonic	GAT GNN	Improvement
Goal reach rate	0.923	0.926	+0.3%
Collision rate	0.050	0.039	-22.0%
Avg. goals reached	22.59	22.67	+0.4%
Avg. collisions	1.34	1.01	-24.6%
Avg. steps	175.1	205.2	+17.2%

Caveats, Difficulties, and Future Directions

Caveats.

- The observed performance gains over the harmonic baseline are primarily attributable to the priority-based repulsion protocol rather than learned policy behavior. The sheaf GNN has not yet converged to a meaningfully differentiated priority assignment for long range corridor deconfliction.

Difficulties.

- Training a residual RL policy on top of a strong base controller produces a weak learning signal — the harmonic navigation system resolves most interactions correctly, leaving little residual for the policy to discover.
- Credit assignment in dense multi-agent settings with sparse rewards made it difficult to reinforce cooperative yielding behavior, as agents that yield receive no direct reward signal for enabling another agent's goal completion.

Future Directions.

- A formal characterization of when the learned sheaf structure converges to a global section — and its relationship to the harmonic Dirichlet energy minimized by the navigation layer — would provide theoretical grounding for the consistency regularization framework.
- Training the full policy end-to-end from motion primitives rather than as a residual correction, with a shaped reward that directly incentivizes successful deconfliction events, would give the RL component a stronger and less ambiguous learning signal.